G H Raisoni International Skill Tech University, Pune

T.Y. B. Tech (All)
(Sem-V)
CAE II- WINTER 2026 (2024 Pattern)
Database Management System (25BTCEPCL501/25BTAIPCL501)

(Time: - 01 Hour) **(Max. Marks: 20)**

Instructions to the students:

i) Solve Q.1 or 2 and Q3 or 4. ii) Neat diagrams must be drawn wherever necessary. iii) Figures to the right indicate full marks. iv) Assume suitable data, if necessary.

Q. No		Question	Marks	CO	BL
Q1. (a)		State and explain any five of Codd's twelve rules that a database system must satisfy to be considered truly relational. Support each rule with a suitable example.	[5]	CO3	2
(b)		Consider the unnormalized relation STUDENT_COURSE (Roll_No, Name, Course_ID, Course_Name, Faculty, Faculty_Dept) with the functional dependencies: Roll_No → Name; Course_ID → Course_Name, Faculty; Faculty → Faculty_Dept. Identify the partial and transitive dependencies present and decompose the relation, step by step, into a set of relations that satisfy 3NF.	[5]	CO3	3
		OR			
Q2.(a)		Explain domain integrity and referential integrity constraints with suitable examples. Also differentiate between a partial functional dependency and a transitive functional dependency.	[5]	CO3	2
(b)		Consider the relation ORDER_DETAILS (Order_ID, Product_ID, Customer_ID, Customer_Name, Product_Name, Price, Quantity) with the functional dependencies: Order_ID, Product_ID → Quantity; Product_ID → Product_Name, Price; Order_ID → Customer_ID; Customer_ID → Customer_Name. Examine whether this relation is in BCNF. If it is not, decompose it into a set of relations that satisfy BCNF, clearly stating the candidate key of each resulting relation.	[5]	CO3	3
Q3.(a)		Explain the ACID properties of a database transaction, illustrating each property with a suitable banking (fund-transfer) example.	[5]	CO4	2
(b)		Consider the schedule S: R1(A); R2(A); W1(A); R1(B); W2(A); W1(B); C1; C2, where Ri(X)/Wi(X) denotes a read/write on item X by transaction Ti and Ci denotes the commit of Ti. Construct the precedence graph for schedule S and determine whether S is conflict serializable. If it is, give an equivalent serial schedule.	[5]	CO4	3
		OR			
Q.4 (a)		Draw and explain the state transition diagram of a database transaction, describing each state briefly.	[5]	CO4	2
(b)		Consider the schedule S: R1(A); W1(A); R2(A); W2(A); R1(B); C2; W1(B); C1, where Ri(X)/Wi(X) denotes a read/write on item X by transaction Ti and Ci denotes the commit of Ti. Determine, with justification, whether schedule S is (i) recoverable, (ii) cascadeless (avoids cascading aborts).	[5]	CO4	3